Comparative analysis of single class anomaly detection models using sensor data of industrial machinery.

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1 Introduction

In this report, a comprehensive analysis and discussion of the Industrial fault detection project has been conducted on the anomaly detection models of the industrial machinery where the analysis of model performance, misclassified samples using visual plot analysis has been presented.

2 Literature review

In this section, literature on the task,metrics, model, and plots has been discussed.

2.1 Predictive maintenance

Industry 4.0 has become a major advancement in the industrialization where more and more industries and factoring are employing the modern technology for manufacturing and maintenance and AI is the most prominent tool that is at the forefront of this major shift. AI in industrial sector has different major applications which can make more efficient progress for example predictive maintenance where the AI model can warn before a machine goes faulty which can save valuable amount of time, and money. AI has shown major advancement in terms of detecting fault accurately. The AI models learn from the historical data from multiple different variables related to the machine and then are able to predict current outcomes of that machine. Another example is RUL(Remaining useful life) is another major application where the AI model predicts the number of remaining cycles in the machine before it goes fault which can minimize or eliminate the downtime given that appropriate action has been taken after the warning from the model for that machine.

2.2 Anomaly detection in industry 4.0

2.3 One class support vector machine

One SVM is a major model in the context of anomaly detection where the model learn the behaviour of a single class for example Normal class of the model where machine is working very well, and when the time for prediction, it classifies all the observations which are not similar to the normal class behaviour. This is alson the baseline model in anomaly detecction because of it simplicity in terms of predictions where the interpretation on the predictions could eb done verz easily.

3 Dataset

The dataset that is used for this project is available in kaggle under the name Equipment monitoring. It has the sensor data of multiple different machines such as Turbine, Pump, and Compressor and has the following different variables

1. Temperature
2. Pressure
3. Humidity
4. Vibration

All the above variables has been normalized for modelling using the sklearn scaling function. The dataset also includes the label column for different observations where 0 is normal class, and 1 is faulty class. Both, Supervised and unsupervised learning could be performed on this dataset and particularly, as there are less faulty observations, there could be a problem with disproportionate classes and resulting in overfitting of normal class as the observation for normal dominate the whole dataset. To cure this problem, Resampling techniques used to add synthetic samples to the minority class , reducing the chance of overfitting to a particular class.

4 Models and metrics

I have fit 2 models based on single class classification, one class SVM model and Isolation tree model and the reason for this application is because in the real world, the proportion of faults compared to normal observations is very low and causes the problem of class imbalance. When it is modeled as supervised, the accuracy could easily be inflated due to model learning the normal class better due to more normal observations and also predicting on normal class better than the faulty class. To do accurate modelling, unsupervised modelling is done for better interpretability and accurate performance. After fitting the models, the following are the results:

4.1 Models

For anomaly detection, One class support vector machine and the isolation forest models have been used which are internally designed for the unsupervised anomaly detection purpose and also very interpretabe with its parameters such as

1. regularization 'nu' parameter of the one class SVM model .
2. Contamination parameter of the isolation forest model.

which are the expected proportion of faulty observations by chance. After training, the model can be tested for test dataset containing both, normal and faulty observations for analysing the model fit and generalization.

4.2 Metrics

The following metrics have been used for performance evaluation of the models and model selection.They are:

1. $F2$ **score:** The overall classification performance of the model.
2. **precision:** Proportion of accurately classified true positives from all the classified true positives.
3. **recall:** Proportion of accurate true positives classifications from actual true positives
4. **Confusion matrix of classifications:** Matrix depicting the number of true and false positives ,true and false negatives
5. **Model size:** Memory footprint of the trained model.
6. **Training time:** Total time taken by the model to complete the training on training dataset
7. **Bootstrapping:** 100 bootstrapping(small) samples created with resampling to analysing the consistency of the model
8. **Hyperparameter tuning:** Tuning of model initalization using a range of values for "nu" hyperparameter of one class SVM and contamination hyperparameter of the isolation forest in the range of 1% to 5% of fault observation proportions in the normal data.

5 Results

In this section, the results for the project ″anomaly detection for industrial machinery″ are discussed. The comparative analysis of both the isolation forest and one class support vector machine models has been done using multiple visual plots, which gives a unique perspective on how the models are performing and their generalization. The analysis has been divided into the following sections:

1. Confusion matrix of models
2. Model selection strategies
3. Anomaly score analysis

5.1 Confusion matrix of models:

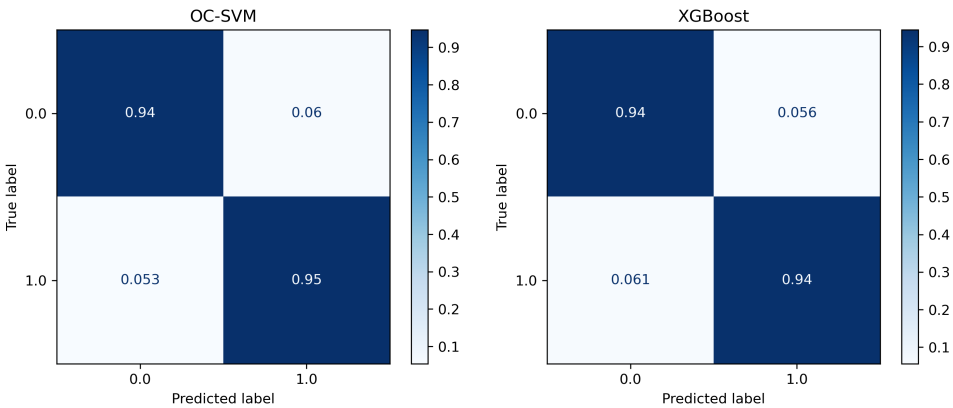
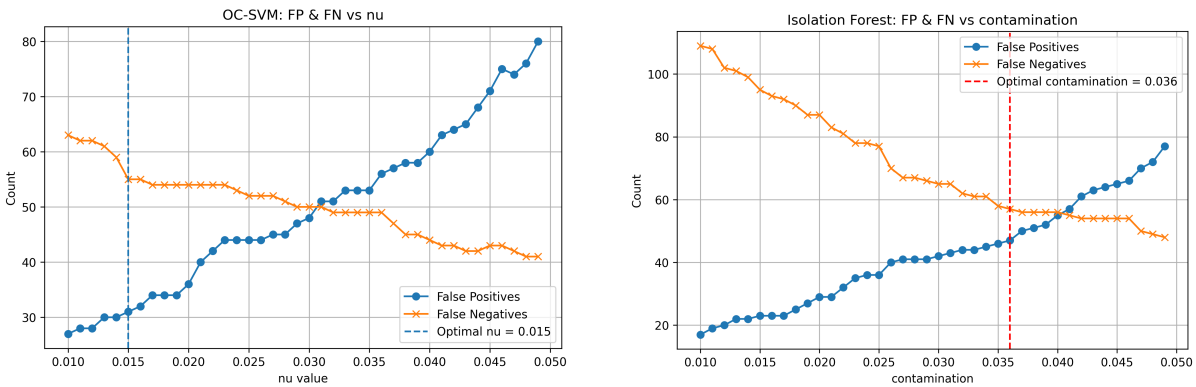


Figure 1: Comparison of fault detection performance between OC-SVM, Isolation Forest.

The above confusion matrices reveal the classification results of both models. As both models, Isolation forest and One class SVM, were trained on normal class, they have learned the underlying features of normal class with 94% and 95% $F2$ scores and were successfully able to differentiate the behavior of faulty sensor observations. Even with some overlapping observational behavior in the sensor readings of both classes, the models are able to discriminate efficiently. As both models are single class based, they seem to be performing similarly with one class SVM slightly better which is classifying 95% of true positives accurately . In the industrial field, detecting faults accurately is desired as it reduces the cost and downtime significantly, One class SVM has shown to better generalize the decision boundary. But the isolation forest has seen to be better at raising lower proportion of false alarms with less than 6% of true positives classified as false alarms. In my view, a model which classifies more and accurate actual faults should be preferred even though there may be slightly more false alarms as former costs significantly more resources if not detected on time than detecting some false one.

5.2 Model selection strategies:

5.2.1 Hyperparameter tuning:



(a) OC-SVM

(b) XGBoost

Figure 2: Normalized confusion matrix comparison between OC-SVM and XGBoost.

In this grid plot, the sensitivity hyperparameter of One class SVM and the contamination hyperparameter of the isolation forest are tuned from 1% to 5% fault proportions in the dataset and the corresponding total classification errors for each value of both hyperparameters are plotted. Both hyperparameters have shown a clear tradeoff between the missed faults and false alarms, but the idea for this analysis is to see fluctuation of model errors with different hyperparameter values. It shows the stability of the models and how well the model is learning the underlying features with different fault proportion assumptions. As there is no sudden change with respect to a specific value of hyperparameter, the models show good generalization and are consistently stable at prediction performance respect to classification errors.

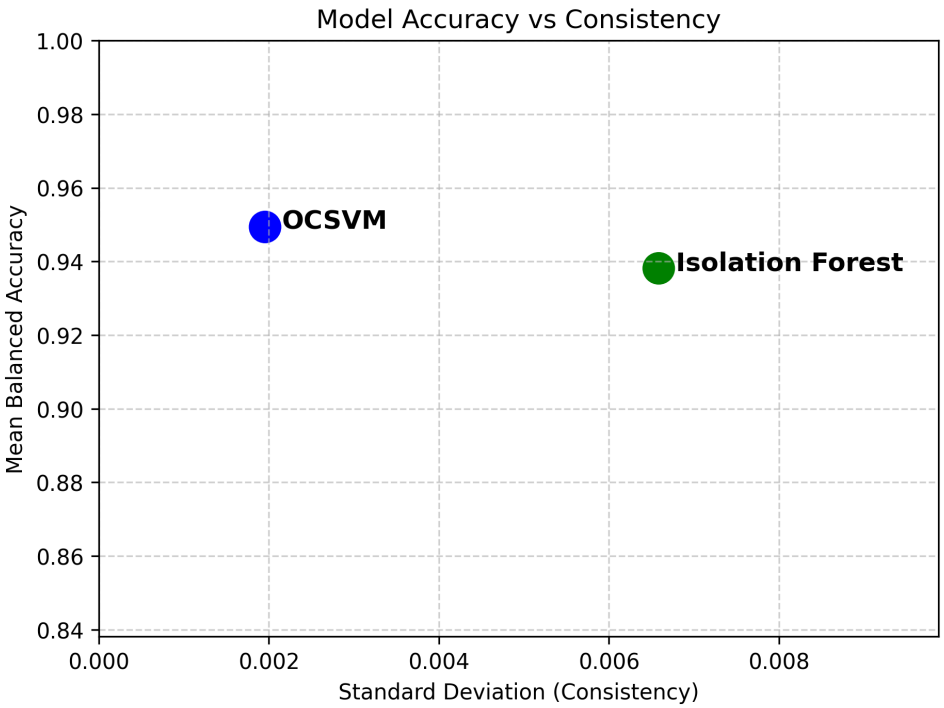


Figure 3: Comparison of fault detection performance between OC-SVM, Isolation Forest, and XGBoost.

The above plot shows the average model accuracy and the variance of among the model accuracies of the sensitivity and contamination hyperparameter tuning. In this, X axis represents the standard deviation showing the consistency with respect to model predictions and Y axis is the average $F2$ score. After all the runs, One class SVM has resulted in the best results with best average accuracy with very low variation among the different values of sensitivity in comparison with isolation forest results.

5.2.2 Multiple model selection parameters:

Model	Bootstrap Recall	Model Size(KB)	Training time(S)
Isolation Forest	0.931	1093.27	0.032985
One-Class SVM	0.999	14.02	0.964153

Figure 4: Comparison of fault detection performance between OC-SVM, Isolation Forest, and XGBoost.

In this subsection, The above table shows multiple parameters that provide an optimistic perspective on the selection of the model. In this table, there are 3 parameters,

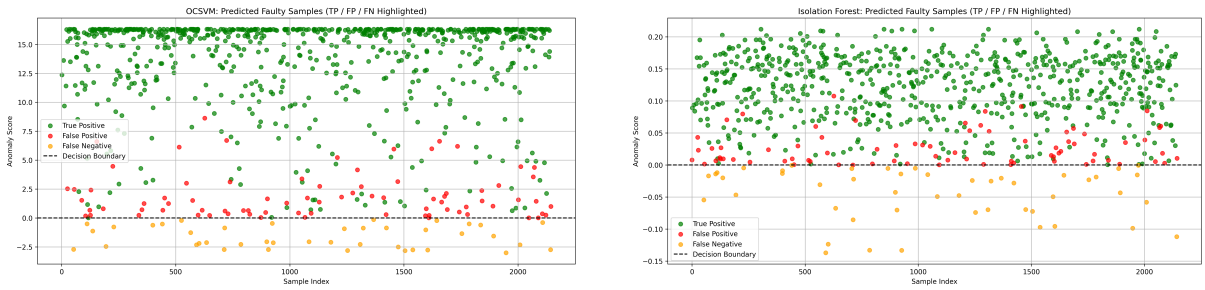
1. **Bootstrap Recall:** Both models have been predicted for 100 bootstrap datasets where One-Class SVM model outperforms the Isolation tree model with significantly better and consistent results.

2. **Model size:** The size of the model plays a huge role in efficient deployment, where a model that has lower memory consumption makes it easy for the deployments and also functions efficiently without memory constraints. One-class SVM has a significantly better memory footprint.

3. **Training Time:** The training time is another factor that could determine the model selection where it depicts how fast the computation is performed. Methodologically, These 2 models differ in terms of structure and computation as One class SVM used RBF kernel, it is essentially fitting the model non linear and have to solve the convex optimization problem for fitting the deterministic boundary for classification where in the Isolation forest, it is random tree scaling which has lower computations than One class SVM. As a result, The Isolation forests trains significantly faster than One-class SVM

Model size can alone be a selection factor when there is a tie in model performance in terms of $F2$ score but as there is a trade off with the training time, It is important to also see the bootstrapping which shows the model robustness. Compared with isolation forest, One class SVM has done exceptionally well with generalization and robustness.

5.3 Anomaly score analysis:

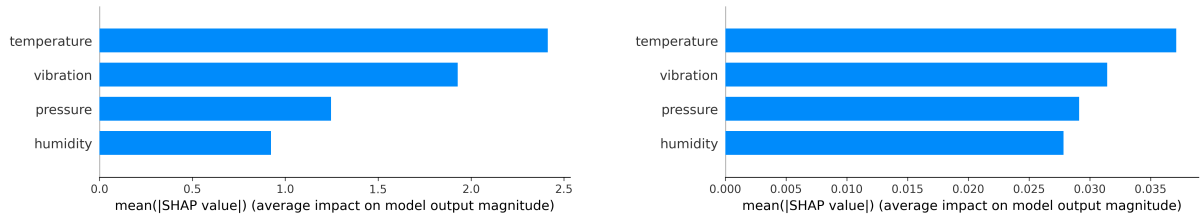


(a) OC-SVM (b) XGBoost

Figure 5: Normalized confusion matrix comparison between OC-SVM and XGBoost.

The above plot depicts the anomaly boundary of each model. This analysis is for the misclassified observations, and all the green dots represent the accurate classification of observation and red dots depict the false negatives. As it is a fault detection model designed for the industrial usage, The focus should be more on accurately detecting the true positives. after analysing the anomaly boundary, the false negatives are short of a small score to be determined as true negatives as they are place extremely close to the decision boundary by the model. This shows that it is very important to consider the anomaly score as an addition parameter to determine the chance of an observation being a true negative. For simpler approach, the point prediction are extremely interpretable but for more efficient classification,It is highly beneficial to also consider the classifications with probabilistic uncertainty estimates for each observation to determine even the misclassified observations.

5.4 Explainer Module:



(a) OC-SVM (b) XGBoost

Figure 6: Normalized confusion matrix comparison between OC-SVM and XGBoost.

An SHAP explainer module has been initialized to provide the classification drivers for either faulty or non faulty observations where the model has been trained for 1000 observations from within test dataset and has provided model agnostic explanations for both models where the top 2 features for both explainer remains the same with almost similar mean SHAP value. Using the above consistency in the feature importance, Based on SHAP analysis,it suggests that these 2 features are the top generic features that drive the observation to be either faulty or non faulty significantly. It is also interesting to investigate model performance which has been fit using only these 2 features or using principle components of these 2 features to see the modeling and classification impact where if it provides a significantly similar performance as the model which has been fit for all the features, then it would be beneficial insight in terms of resource optimization as in deployment of less sensors to get the similar kind of performance.

6 Conclusion

In conclusion, the application of AI shows a promising potential in make the industrial operations more efficient and consistent. As shown in the above analysis of the project, Fault detection is one of the many application of AI where the model can provide an operational evolution to the whole production pipeline. Incorporation of AI mode as a detection model to accurately detect faults would return a huge beneficial efficiency in terms of cost, downtime, and smooth workflow of operations. In comparison with tree based isolation forest, One class support vector machine has demonstrated a better return on investment of resources performance in terms of input resources. There still limitations with some observations being not accurately classified but for a better classification, a better quality dataset is desired with more driving features.